

WASP-140b

R-2159

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_211202 · created 2026-08-12T23:36:10+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459550.818 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.012
Transit depth (Rp/Rs)^2	2.97 +/- 0.4 [%]
Orbital Inclination (inc)	83.64 +/- 0.33
Airmass coefficient 1 (a1)	1.3285 +/- 0.0095
Airmass coefficient 2 (a2)	-0.1395 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	None
Optimal Aperture	5.03
Optimal Annulus	5.87
Transit Duration (day)	0.0699 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.012 vs 0.1432 (deviation 2.22σ)
Duration fit vs published	0.0699 d vs 0.0604 d (deviation 1.86σ)
Mid-time O–C	6.5 min
Depth SNR	13.2
Residual scatter	0.71 %

Light curve

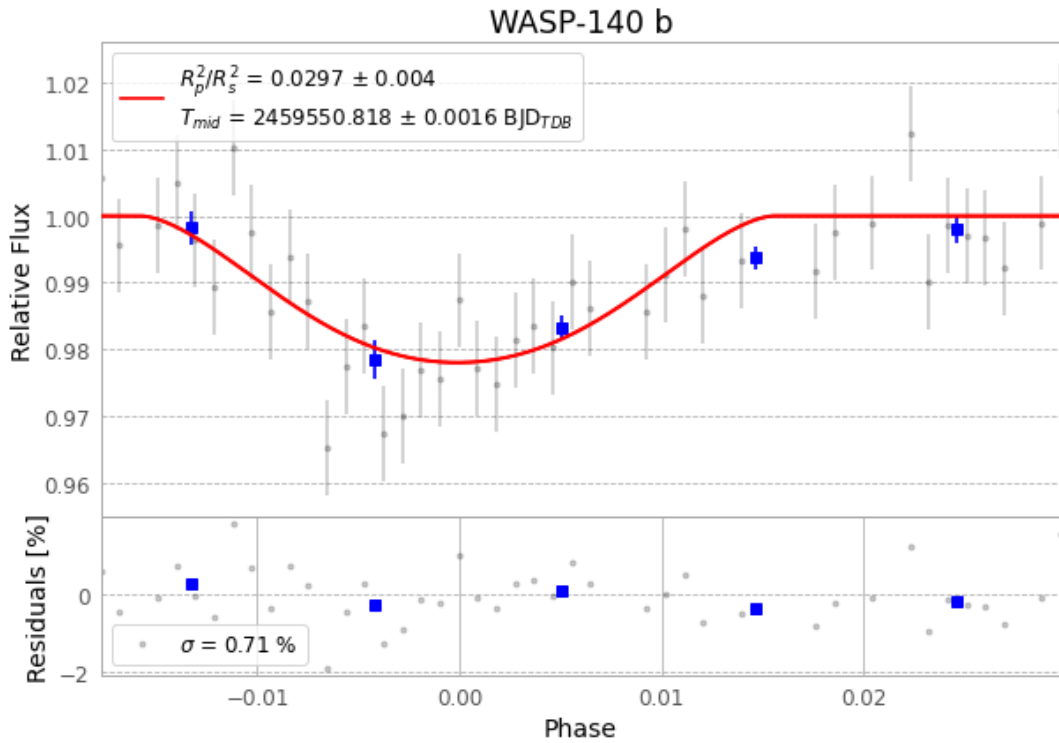


Figure 1 — FinalLightCurve_WASP-140 b_2021-12-01.png (SHA-256 7d0acc571cdbbba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 123], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d29376994b9dbffb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

43 FITS frames (28 MB), WASP-140211202063312.FITS → WASP-140211202090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-211202.log (e1ecf4f54878...), FinalLightCurve_WASP-140 b_2021-12-01.png (7d0acc571cdb...), FinalLightCurve_WASP-140 b_2021-12-01.csv (803852c1c390...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:36Z.